

VAIBHAV CHHABRA

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SUMMARY

Senior Security Engineer with 10+ years of experience designing cloud-native detection systems, security automation, and telemetry-driven analytics across Linux, AWS, and Kubernetes environments. Experienced building adversary emulation frameworks, scalable detection pipelines, and production security tooling using Python, Docker, Kubernetes, and cloud telemetry. Published researcher with expertise in behavioral anomaly detection, distributed systems, and applied machine learning.

SKILLS

Security Engineering & Detection Infrastructure: Linux internals, cloud-native detection pipelines, container observability, Kubernetes, Amazon Elastic Kubernetes Service (EKS), telemetry correlation, detection engineering, real-time alerting, detection triage automation

Cloud & Platform Engineering: AWS, Docker, Kubernetes, Git, CI/CD, REST APIs, cloud-native architectures, distributed systems, containerized testbed orchestration

Programming & Automation: Python, Shell, PowerShell, C/C++, workflow orchestration, detection automation, log parsing

Threat Research & Detection Engineering: Adversary behavior modeling, MITRE ATT&CK, Sigma, Falco, Procmon, Endpoint Detection and Response (EDR), Extended Detection and Response (XDR), OPSEC evasion analysis, IOC clustering, behavioral analytics

Machine Learning & AI: Scikit-learn, Pandas, NumPy, PCA, UMAP, HDBSCAN, RBMs, VAEs, anomaly detection, clustering, drift analysis, statistical analysis

Malware Analytics: Static malware analysis, malware telemetry analysis, Portable Executable (PE) analysis, VirusTotal automation, YARA integration, exploit proof-of-concept validation, protocol analysis

Visualization & Communication: Kibana, interactive dashboards, technical reporting

EXPERIENCE

Drift Analytics LLC | Jul 2025 – Present

Senior Security & ML Engineer

- Built daily Python telemetry pipelines using Scikit-learn, UMAP, and HDBSCAN to ingest and extract features from distributed network data.
- Engineered unsupervised ML models (CDAEs, RBMs, VAEs) for temporal drift analysis and latent space anomaly detection without labeled training data.
- Published peer-reviewed research on structural anomaly detection and latent geometry ([arXiv:2605.20391](#)) and authored U.S. Provisional Patent (App. No. 64/034,500).
- Applied behavioral telemetry analytics and representation learning to evaluate anomaly detection frameworks for autonomous AI systems and distributed networks.

Trend Micro, Austin, TX | Mar 2022 – Jun 2025

Threat Researcher

- Researched Linux container and cloud attack techniques, including container escape, privilege escalation, and fileless persistence.
- Built Kubernetes (Amazon Elastic Kubernetes Service (EKS)) adversary emulation environments to validate cloud-native detections and generate production telemetry.
- Developed Python automation and orchestration workflows to improve detection validation and engineering efficiency.
- Correlated Linux `auditd`, AWS CloudTrail, Kubernetes Audit Logs, and XDR telemetry to improve detection fidelity and reduce false positives.
- Authored Sigma detections mapped to MITRE ATT&CK and integrated them into Trend Micro XDR.
- Partnered with detection and engineering teams to close telemetry gaps and improve production detection coverage.

Trend Micro, Austin, TX | Sep 2013 – Mar 2022

Security Researcher

- Developed high-performance Intrusion Prevention System (IPS) signatures to detect vulnerability exploitation and protocol abuse.
- Reproduced exploit proof-of-concepts and malware behavior in lab environments to develop production detection logic.
- Developed tooling integrating VirusTotal, YARA, Portable Executable (PE) analysis, and internal telemetry to accelerate malware triage and signature development.
- Investigated malware behavior using heuristic analysis and similarity-based approaches to support detection development.
- Delivered production detection content protecting enterprise environments in real time.

PATENTS

- *Network Traffic Pattern Based Machine Readable Instruction Identification*
U.S. Patent No. [US10757029B2](#)
- *System and Method for Geometric Manifold-Based Anomaly Detection in Networks*
U.S. Provisional App. No. 64/034,500 (Patent Pending)

PUBLICATIONS

- Chhabra, V. (2026). Latent Geometry as a Structural Monitor. [arXiv:2605.20391](#)
- Chhabra, V. (2026). Drift as Deformation. [Zenodo: 18463659](#)

EDUCATION

University of Oklahoma – BS, Electrical Engineering (Minor: Physics)

University of Miami – MS, Music Engineering Technology

- Applied wavelet packet decomposition for scalable audio feature extraction.